

C++ Programming Language

Bitwise Operators

Binary Number System | Decimal Number
01 | 0123456789

$$\text{int num1} = (27)_{10} = (11011)_2$$

Decimal → Binary

2	27	1
2	13	1
2	6	0
2	3	1
2	1	1
	0	

Binary → Decimal

$$\begin{array}{ccccccc} & 1 & 1 & 0 & 1 & 1 & \\ 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & & \\ 16 & 8 & 0 & 2 & 1 & & \\ \hline 16 & + & 8 & + & 0 & + & 2 & + & 1 & = & 27 \end{array}$$

* 16, 22, 42
47, 59, 21

$$(39) \rightarrow 100111$$

64	32	16	8	4	2	1
	1	0	0	1	1	1

0
MSB

1
MSB

$$42 \rightarrow 101010$$

64	32	16	8	4	2	1
	1	0	1	0	1	0

Symbol	Name
&	AND
	OR
^	Ex-OR
~	NOT
<<	Left shift
>>	right shift

A	B	A & B	A B	A ^ B
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

$$\begin{array}{r} 0100 \\ \& 1000 \\ \hline 0000 \end{array} \quad \begin{array}{r} 0100 \\ | 1000 \\ \hline 1100 = 12 \end{array} \quad \begin{array}{r} 0100 \\ \wedge 1000 \\ \hline 1100 = 12 \end{array}$$

$$\begin{array}{l} 0 \rightarrow 1 \\ 1 \rightarrow 0 \end{array}$$

$$\sim n = -(n+1)$$

$$\sim 4 = -(4+1)$$

$$\sim 7 = -8$$

$$\sim 8 = -9$$

$$\begin{array}{r} 0000 \\ + 1 \\ \hline 1111 \end{array}$$

Two's Complement

Decimal → Binary

void main() {

int decimal = 27;

String binary = decimal.
toRadixString(2);

print(binary);

}

Binary → Decimal

void main() {

String binary = '11011';

int decimal = Int.parse(binary,
radix: 2);

print(decimal);

}

void main() {

int a = 4;

int b = 8;

print(a & b); // 0

print(a | b); // 12

print(a ^ b); // 12

}

4 = 100;

8 = 1000;

D	B
00	00
01	01
02	10
03	11
04	100
05	101
06	110
07	111
08	1000
09	1001
10	1010

$$\begin{array}{r} 100 \\ 011 \\ \hline -5 \end{array}$$

$$\begin{array}{r} 0101 \\ 1010 \\ + 1 \\ \hline 1111 \end{array}$$

$$111111111011$$

<< → Left shift

```
void main() {
```

```
    int a = 4;
```

```
    int i = 2;
```

```
    print(a << i);
```

```
} // 16
```

100
1 0 0 0
16 8 4 2 1

>> → right shift

```
void main() {
```

```
    int a = 8;
```

```
    int i = 1;
```

```
    print(a >> i);
```

```
}
```

USES

1) Fast

odd/Even

```
void main() {
```

```
    int a = 9;
```

```
    if (a % 2 == 0) {
```

```
        print(a is Even);
```

```
    }
```

```
    else
```

```
    { print(a is odd );
```

```
    }
```

```
}
```

a = 8

1000
0001
0000

1 → odd

0 → Even

MSB

```
void main() {
```

```
    int a = 9;
```

```
    if (a & 1 == 0) {
```

```
        print(a is Even);
```

```
    }
```

```
    else
```

```
    { print(a is odd);
```

```
    }
```

```
}
```

2¹ = 2

2² = 4

2³ = 8

2⁴ = 16

```
void main() {
```

```
    int a = 32;
```

```
    if (a > 0 && (a & (a - 1) == 0)) {
```

```
        print(a is power of 2);
```

```
    }
```

```
    else {
```

```
        print(a is not power of 2);
```

```
    }
```

1 0 0 0 0 0 → 32
32 16 8 4 2 1

1 1 1 1 1 → 31

100000
011111
000000

1) 4 & 6

2) 8 1 2

3) 7 ^ 3

4) 11 << 2

5) 13 >> 3

H/w